

BLAST FURNACE PRODUCTION SECTION												
DATE: 1. Feb. 2025		DAILY PARAMETERS REPORT										
Monthly Plan (T)		ABP 541000	OPP 450000	Sinter Prod. / Nut Coke		EB 6950	0					
Daily Rate (T/Day)		17006		SP - III 8950		51						
H.M.SENT TO SMS's		16935	COAL STOCK Dry Coke ==>		295636 M/c-II 7244 TOTAL	9795	51					
						25695						

HOT METAL AND SLAG QUALITY

Parameter	BF-1	BF-4	BF-5	BF-6	BF-7	BF-8	SHOP	TILL	(BF-4 to 7)
Avg. % 'Si'		0.84		0.53	0.82	0.63	0.69	0.69	0.83
Min/Max		0.59/0.95		0.29/1.02	0.58/1.13	0.49/1.17			
Till % 'Si'		0.84	#DIV/0!	0.53	0.82	0.63			0.83
Avg. % 'S'		0.017		0.033	0.018	0.034	0.028	0.028	
Min/Max		0.010/0.027		0.010/0.055	0.010/0.026	0.020/0.047			
Avg. % MgO		7.65		7.40	6.19	7.87	7.28		
Min/Max		7.13/8.37		6.61/8.12	5.66/7.04	7.16/8.34			
Avg. % Al2O3		16.75		16.92	18.05	17.76	17.37		
Avg. % FeO		0.54		0.63	0.6	0.89	0.67		
Avg. % K ₂ O		0.38		0.33	0.31	0.32	0.34		
BASICITY(-)		1.04		1	1.07	0.98	1.02		
Min/Max		0.98/1.11		0.94/1.06	0.97/1.12	0.92/1.01			
Avg. % 'P'		0.17		0.12	0.13	0.10	0.13		
Min/Max		0.16/0.19		0.11/0.14	0.12/0.15	0.10/0.11			

Fce.No.	Si→1.00	Si→0.85	No.of Casts	Tap to Tap >1.00 Hrs
4	0	1	6	3
5			0	
6	1	2	7	1
7	1	4	7	0
SHOP	2	7	20	4

H ₂ IN TOP GAS						
	BF-1	BF-4	BF-5	BF-6	BF-7	BF-8
G		4.50	STOP	4.90	3.20	
CO/CO ₂ RATIO						
G		1.20		1.26	1.25	
% Gas Util.						
G		45.55		44.29	44.5	44.28

MIXER	%Si	%Mn	%Sul	%P
Mixer-III				
Mixer-IV				

PELLET CHEMICAL AND SEIVE ANALYSIS

	Fe	SiO ₂	Al ₂ O ₃	CaO	MgO	+ 10 MM	- 5 MM	M.Size	Basicity	Slag Comp.
BF # 8 =>	61.64	7.70	3.84	1.60	1.71	56.00	2.70	11.00	0.21	14.85
BF # 7 =>	60.98	7.06	4.1	1.55	1.51				0.22	14.22
BF # 5 =>									#DIV/0!	0.00
BF # 4 =>									#DIV/0!	0.00

SKIP IRON ORE				
	Fe (%)	SiO ₂ (%)	Al ₂ O ₃ (%)	Moist.
BF # 4	64.09	3.03	2.74	4.2
BF # 5	STOP			
BF # 6	63.71	3.58	2.65	4
BF # 7	62.63	4.36	3.46	3.8
BF # 8	62.37	5.80	2.50	3.6
Avg.	63.20	3.35	2.27	3.9

BF SKIP I/ORE SEIVE ANALYSIS			
Fce.No.	+40 mm	- 10 mm	M.Size
BF # 4	24.73	11.57	28.27
BF # 5			
BF # 6	22.55	9.70	27.95
BF # 7	22.85	8.95	28.90
BF # 8	20.27	13.13	26.88
AVG.	22.60	10.84	28.00

% FINES IN BF SKIP SINTER				TOTAL Fe			% FINES IN BF SKIP COKE			
Fce.No.	-10 mm	- 5 mm	M.Size	SHIFT A	SHIFT B	SHIFT C	Fce.No.	-40 mm	- 25 mm	M.Size
BF # 4	30.97	6.93	14.80	49.39	49.18	49.79	BF # 4	32.23	1.70	52.00
BF # 5							BF # 5			
BF # 6	29.40	6.40	14.85	50.27	50.70		BF # 6	29.05	1.40	53.20
BF # 7	28.90	5.75	15.40	49.19	49.09	53.15	BF # 7	26.95	1.65	53.30
BF # 8	29.42	3.48	15.55	49.32	49.54	51.13	BF # 8	41.35	2.15	43.25
AVG.	29.67	5.64	15.15				AVG.	32.40	1.73	50.44

COKE QUALITY									
PROXIMATE ANALYSIS					COLD STRENGTH			HOT STRENGTH	
	Moisture	Ash	V.M	F.C	M-40	M-10	Sulphur	CSR	CRI
CSP-I: A	4.1	14.4	1.0	84.2			0.57		
CSP-II: A	4.3	15.1	0.9	83.5			0.59		
CSP-III: A	5.3	14.6	0.9	84.0	80.6	7.6	0.58	64.1	23.9
CSP-IV: A	0.3	14.6	0.9	84.1	85.6	7.2	0.59		
BF # 8 STOCK HOUSE SURFACE MIX.COKE ==>								65.0	23.5

COAL BLEND (%)													
	DATE	PCC	MCC	Aus(H)	BENGA (H)	LVHR (H)	TUHUUP(hard)	Russian (S)	AUS(S)	USA(H)	USA(S)	% IMP	ASH %
CHP - I	1-Feb-25	12.0	3.0	34.0	5.0	12.0	0.0	0.0	17.0	12.0	5.0	85.0	CHP - I 10.0
CPP - II	1-Feb-25	12.0	3.0	36.0	5.0	10.0	0.0	0.0	16.0	12.0	6.0	85.0	CPP - II 11.1

CHEMICAL ANALYSIS

SINTER PLANT-2															
NORM(Basicity)		(1.90 - 2.00) (09-12-24) Nut Coke					7 Kg/T of Sinter								
NORM(Mgo)		(2.75 - 3.25) (05-07-24) MILL SCALE					29 Kg/T of Sinter								
SAMPLE	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Component				
AO	50.19	9.03	8.29	2.74	14.29	3.08	1.72	6.00	OUT	1.72	28.79				
A1	49.25	9.03	8.62	2.66	14.99	3.13	1.74	6.37	OUT	1.74	C1 (SP # II)				
A2	49.74	9.03	8.33	2.68	14.54	3.26	1.75	6.21	OUT	1.75	Coke	Flux	DTI		
BO	49.13	9.00	8.79	2.12	15.18	3.17	1.73	6.39	OUT	1.73					
B1	50.19	9.00	8.60	2.06	14.82	3.04	1.72	6.22	OUT	1.72					
B2	49.91	9.00	8.29	2.49	14.45	2.82	1.74	6.16	OUT	1.74					
CO	49.85	9.00	8.58	2.45	15.08	2.95	1.76	6.50	OUT	1.76					
C1	49.94	9.00	8.27	2.76	14.49	2.95	1.75	6.22	OUT	1.75					
C2	49.86	9.00	8.70	2.58	14.78	3.05	1.70	6.08	OUT	1.70					
DAY AVG.	49.78	9.01	8.50	2.50	14.74	3.05	1.73	6.24	OUT	1.73					

SINTER PLANT-3													
NORM(Basicity)		(1.90 - 2.00) (09-12-24) Nut Coke					0 Kg/T of Sinter						
NORM(Mgo)		(2.75 - 3.25) (05-07-24) MILL SCALE					11 Kg/T of Sinter						
SAMPLE	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Component		
AO	51.28	9.16	8.08	2.05	14.08	2.68	1.74	6.00	OUT	1.74	27.48		DTI
A1	50.61	9.16	8.35	2.30	14.27	2.90	1.71	5.92	OUT	1.71			
A2	51.47	9.16	7.76	2.25	13.65	2.91	1.76	5.89	OUT	1.76			
BO	50.45	9.22	8.30	2.35	14.17	3.28	1.71	5.87	OUT	1.71			
B1	50.94	9.22	8.05	2.13	14.08	3.10	1.75	6.03	OUT	1.75			
B2	51.21	9.22	8.05	2.36	13.78	2.78	1.71	5.73	OUT	1.71			
CO	49.56	9.22	8.51	2.49	15.14	2.87	1.78	6.63	OUT	1.78			
C1	50.77	9.22	8.20	2.43	14.07	2.85	1.72	5.87	OUT	1.72			
C2	51.05	9.22	8.13	2.04	14.12	2.74	1.74	5.99	OUT	1.74			
DAY AVG.	50.82	9.20	8.16	2.27	14.15	2.90	1.74	5.99	OUT	1.73	50.45		
											27.56		
											50.79		
Sample ID	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Comp		DTI
SP-3/S/MC/2A0	50.85	9.58	8.02	2.79	13.91	2.79	1.74	5.89	OUT	1.73	27.64		
SP-3/S/MC/2A1	50.39	9.58	8.25	2.58	14.54	2.95	1.76	6.29	OUT	1.76			
SP-3/S/MC/2A2	50.28	9.58	8.37	2.72	14.43	3.01	1.72	6.06	OUT	1.72	C I (SP # III)		
SP-3/S/MC-2/B0	51.34	9.65	8.07	1.92	13.70	2.86	1.70	5.63	OUT	1.70	Coke	Flux	
SP-3/S/MC-2/B1	51.87	9.65	7.65	2.47	13.20	2.96	1.73	5.55	OUT	1.73			
SP-3/S/MC-2/B2	51.31	9.65	7.94	2.24	13.66	2.89	1.72	5.72	OUT	1.72			
SP-3/S/MC-2/C0	50.37	9.65	7.97	2.72	14.26	2.95	1.79	6.29	OUT	1.79			
SP-3/S/MC-2/C1	49.85	9.65	8.42	2.81	14.57	3.23	1.73	6.15	OUT	1.73			
SP-3/S/MC-2/C2	50.54	9.65	8.32	2.50	14.21	2.87	1.71	5.89	OUT	1.71			
DAY AVG.	50.76	9.63	8.11	2.53	14.05	2.95	1.73	5.94	OUT	1.73			

BLAST FURNACE DEPARTMENT

Stoppage, L.Blast, Reduced Blast & offrod Duration and Reasons for Dated

1-2-25

on Dtd

2-Feb-2025

8:55

	Duration		REASONS	ST	LB	RB	FB	Down Time	Furnace No	1	4	5	6	7	8	TOTAL	
I			== FCE. BLOWN OUT ==						ABP	0	70000	70000	71000	108000	222000	541000	
									OPP	0	69700	0	73700	106300	200300	450000	
IV	06:00	11:05	RT. COKE GATE OPENING PROB. & ROLL BACK PROB.						M/Rate	0	61992	0	72380	112056	229740	476168	
			FCE. LEVEL DOWN 4.5 Mtr.						Weekly Plan	0	15250	0	15900	10800	48000	89950	
	10:00	12:05	CABLE FAULT (ELECT.)					Prod.(Cumm.)	0	13581	0	14885	11162	49172	88800		
			C7 BELT GRAVITY TAKE UP DRIVE SIDE DEFLECTION DRUM					Fulfillment	#DIV/0!	89.1	#DIV/0!	93.6	103.4	102.4	98.7		
			SHAFT BROKEN, SLAG IN LADLE					Today'sPlan	0	2550	2550	2600	3900	8100	19700		
			J5 BUNKER FLAP GATE MOTOR BRACK DOWN					HM Prod	0	2214	0	2585	4002	8205	17006		
13:45	14:00	DUST CATCHER CLOSING PROB.					Cu.Prod	0	2214	0	2585	4002	8205	17006			
								24:00:00	D/Rate	0	2214	0	2585	4002	8205	17006	
V	06:00	05:59	SCHD.MAINT.	23:59					No.OF SKIPS (6:00 TO 6:00)							Charge	Total
				23:59				0:01:00	A		186		198	273	46	657	
VI	06:00	12:00	LSV CLOSING PROB., LEVEL DOWN 4.3 Mtr.					23:59:00	B		201		223	254	47	678	
	12:00	13:00	==DO==			1:00			C		197		214	263	44	674	
	14:00	16:00	LOOSE CONTACT (ELECT. PROB.)			2:00			Total skips	0	584	0	635	790	137	2009	
	19:30	21:10	P.R.M.S PROB., OXY ZERO, (INSTRUMENTATION. PROB.)						Stoppage on Date			23:59				23:59:00	
	20:05		BLOWER-07 CONNECTED						Low Blast on Dt					1:00		1:00:00	
						3:00	21:00:00	0:45:00	Reduced blast on Dt				3:00	1:00		4:00:00	
VII			C7 BELT GRAVITY TAKE UP DRIVE SIDE DEFLECTION DRUM						No of Tappings	0	6	0	7	7	12	32	
			SHAFT BROKEN					No of Not Dry casts	0	0	0	0		0	0		
	19:05	20:05	BLOWER-11 TRIP AT 19:05, BLOWER-12 CONNECTED AT 19:30		1:00			Coke Rate On Date		452		457	459	393	426		
			DUE TO BLOWER TRIP			1:00		Coke Rt for month		452		457	459	393	426		
	20:00	21:00				1:00	22:00:00	0:45:00	Hot Blast Temp.		1001		1003	941.45833	1152	1049	
	VIII			==NIL==						Metal ladle position				Slag ladle position			
								24:00:00	0:00:00	In circulation	36	+80	2	Total	In Use	Under M/Maint.	Under E/Maint.
			TOTAL	23:59:00	1:00:00	4:00:00	91:01:00	25:29:00	Open Hearth		+75	4					

COAL DUST INI. (Kg/THM)				
BF		CDI		
No.	DAILY/ Dat	Rate(kg/thm	HM Production on that date(T)	CDI in tonnage(T)
1	25.03.2021	124	1852	230.15
4	14.12.2022	226	1250	281.90
5	03.11.2024	209	1118	233.62
6	10.10.2022	206	1821	313.14
7	07.08.2024	158	3083	488.15
8	05.11.22	192	7651	1472.00
SHOP	21.10.24	160	14750	2360.00

CDI On Tonnage Basis			
BF			
No.	DAILY	Best CDI On tonnage Basis(T)	HM Production on that date(T)
1	25.03.2021	230.15	1852
4	18.10.2024	410.99	1954
5	07.11.2024	427.53	2304
6	19.09.2022	334.99	2044
7	10.12.2022	540.30	4176
8	30.11.2022	1575.97	8793
SHOP	25.01.2023	2721.79	18771

Annual ABP Target(FY 2024-25)		Till Date
HM Target(MT)	6.60	4.89
Coke RATE(kg/thm)	429	432
CDI Rate(kg/thm)	133	137
FUEL RATE(kg/thm)	588	590
Financial Data		
Cost of HM Production in FY 2023-24 (per ton of HM)	Rs 35524	
Cost of HM Production in FY 24-25 till Oct'24(per ton of HM)	Rs 33554	
Cost of HM Production in Oct'24(per ton of HM)	Rs 32291	
Contribution Margin per ton (Profit) for Pig Iron in oct-24	Rs 8995	
Cost of Skip coke in FY 24-25 Till Oct'24 (per ton)	Rs 38175	
Cost of Skip Coke in Oct'24(per ton)	Rs 36476	
Previous Year ABP vs Actual (FY2023-24)	ABP	ACTUAL
HOT METAL(MT)	6.20	5.99
COKE RATE(kg/thm)	430	455
CDI RATE(kg/thm)	125	120
Fuel Rate(kg/thm)	587	597
Pig Iron Produced in FY 2023-24	67695 T	
Pig Iron Produced in FY 2024-25	26673 T	

Metal ladle position		Slag ladle position	
In circulation	36	+80	2
Open Hearth		+75	4
Reserve	29	+70	13
U/Repair	4	-70	12
Cooling	3	Total	31
Mixer Position & M- III		% Screen Loss	
Pouring AT M-IV			
6:00:00 AM SMS-III (TPD)			
%SINTER FINES		SP # II	SP # III
		30.87	21.10
Coal Dust Injection		BF#1	BF#4
Coal Consum.(T)		0.00	281.00
Coal Rate KG/THM		#DIV/0!	127
Coal Prog. KG/THM		0	17
Raw Mat. Position			
(FROM RMD)	L/STONE	Mn.Ore	Qz.
Receipt	0	0	0
Stock	12133	48	3394

RECORDS					
B.Fces.	Hot Metal Production		COKE RATE		CDI RATE
	MONTHLY		MONTHLY		MONTHLY
BF # 1	61574	AUG.'17	436	MAR.'12	124
BF # 4	82493	MAY.'24	438	MAR.'23	154
BF # 5	82758	MAR.'22	418	FEB.'09	162
BF # 6	89535	JAN.'09	397	JAN.'02	144
BF # 7	128247	OCT.'17	428	MAR.'19	124
BF # 8	252731	DEC.'24	370	SEP.'21	160
SHOP	555571	DEC.'24	414	SEP.'21	150
PCM POURING	41	2451	30-06-2023		